

40 Solved MCQs

Every answer worked out.

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The circled option is the answer. The boxed working below it is how you would get there in the exam.

Q1–Q3 are actual questions from sir's own End Term Pattern slides, with his marking: **+4 for right, –1 for wrong**. Three more questions in this set are built on those same patterns, because he reuses them.

Five questions are from the guest-lecture session — sir has said that block is worth 4–5 questions.

Q1

★ ACTUAL PYQ · Terminal value

The EBITDA of a company in the last year of the explicit forecasting period is \$200 million. The free cash flow for the same year is \$80 million. The cost of capital is 12%. If the terminal growth rate is 5%, what is the terminal EBITDA multiple?

A 8.6x

B 6.0x

C 5.7x

D 15.0x

Ans: B

WORKING

$$\begin{aligned} \text{TV} &= \text{FCF} \times (1+g) \div (\text{WACC} - g) = 80 \times 1.05 \div (0.12 - 0.05) \\ &= 84 \div 0.07 = \$1,200\text{m} \end{aligned}$$

$$\text{Terminal EBITDA multiple} = \text{TV} \div \text{EBITDA} = 1,200 \div 200 = 6.0\times$$

⚠ 5.7x is what you get by dropping the (1+g): $80/0.07 = 1,142.9 \div 200$. 15.0x divides by the FCF instead of the EBITDA.

This is the multiplier consistency check in reverse: your Gordon terminal value implies a multiple, and 6.0x is what a mature, modestly growing business should look like. If it had come out at 25x, the g or the WACC would be wrong.

Q2

★ ACTUAL PYQ · VC method

After Series A the cap table is: founders 1,000,000 shares, Series A 250,000 shares (total 1,250,000). The Series A investor holds 20%. At Series B the pre-money is \$50 million and a new investor puts in \$20 million. If Series A wants to hold 20% after Series B, approximately how much must it invest in Series B?

- A \$14 million
- B \$4 million
- C \$3 million
- D **\$5 million**

Ans: D

WORKING

Price per share = $\$50\text{m} \div 1.25\text{m shares} = \40.00

Series A's existing stake = $250,000 \times 40 = \$10\text{m}$

Let x = what Series A puts in. Post-money = $50 + 20 + x = 70 + x$

$$(10 + x) \div (70 + x) = 0.20$$

$$10 + x = 14 + 0.2x \implies 0.8x = 4 \implies x = \$5\text{m}$$

⚠ **\$4m is the trap: 20% of the \$70m post-money is \$14m, less the \$10m already held. That ignores the fact that your own investment enlarges the post-money too. You must solve for x on both sides.**

Check: A buys $5 \div 40 = 125,000$ shares $\rightarrow$ holds 375,000 of $1,250,000 + 500,000 + 125,000 = 1,875,000 = 20.0\% \checkmark$

Q3

★ ACTUAL PYQ · Quant

An analyst regressed the P/E of 10 companies in a sector on their projected growth rate in EPS. Figures in brackets are t-statistics:

$$\text{P/E} = 16.38 + 1.2 \times g(\text{EPS})$$

(3.24) (0.8)

Which is the most likely interpretation?

- A The ROE of an average company in the industry is lower than the required rate of return
- B Companies with a higher projected growth in EPS do necessarily trade at a higher P/E
- C **None of the above**
- D Factors other than growth in EPS have no impact on the P/E of these companies

Ans: C

WORKING

$n = 10, k = 2 \Rightarrow df = 8, 5\% \text{ critical } t \approx 2.31$

intercept $t = 3.24 \Rightarrow \text{significant}$

slope $t = 0.8 \Rightarrow \text{NOT significant}$ – growth does not explain P/E in this sample

Take the options one at a time. The growth option is contradicted by the insignificant slope. "Other factors have no impact" overclaims — an insignificant growth coefficient says growth does not explain P/E, not that nothing else does. The ROE statement is not addressable from this regression at all. So **none of the above**.

⚠ The whole question turns on reading the bracketed numbers as t-statistics, not standard errors. Rule of thumb $|t| > 2$; with only 10 observations the bar is higher still at 2.31.

Q4

Multiples

A company earns an ROE of 15%, has a cost of equity of 15%, and grows at 6% a year. What is its justified price-to-book ratio?

- A 1.5x
- B 2.5x
- C Cannot be found without the payout ratio
- D 1.0x

Ans: D

WORKING

$$P/B = (ROE - g) \div (K_e - g) = (0.15 - 0.06) \div (0.15 - 0.06) = 1.0x$$

When $ROE = K_e$ the g cancels — P/B is 1.0 whatever the growth rate. The company earns exactly what its shareholders demand, so growth creates nothing ($NPVGO = 0$).

Q5

Multiples

A firm is expected to grow at 0% and trades at a P/E of 12.5x. What is its cost of equity?

- A **8.0%**
- B 12.5%
- C 6.25%
- D Cannot be found without ROE

Ans: A

WORKING

At $g = 0$: $P/E = 1 \div K_e \Rightarrow K_e = 1 \div 12.5 = 8.0\%$

△ The ROE term vanishes at zero growth — which is exactly why P/E tells you nothing about ROE there, and P/B does.

Q6

Multiples

Company A trades at 15x with 10% growth; Company B at 24x with 20% growth. Which looks cheaper on PEG, and what is the objection to relying on it?

- A A at 1.5 vs B at 1.2 — PEG always favours low growth
- B **B at 1.2 vs A at 1.5 — but PEG assumes P/E is linear in g and ignores risk entirely**
- C They are identical once growth is netted off
- D PEG cannot be compared across different growth rates

Ans: B

WORKING

$PEG_A = 15 \div 10 = 1.5$ $PEG_B = 24 \div 20 = 1.2 \Rightarrow B$ looks cheaper

But P/E is **convex** in g — g sits in the numerator through the payout term and in the denominator through $(K_e - g)$. So dividing by g systematically flatters fast growers. PEG also carries no risk term: same PEG, very different betas, not equally attractive.

Q7

Multiples

Reported PAT is ₹120 cr, which includes a ₹30 cr post-tax gain on the sale of surplus land. There are 10 cr shares and the price is ₹250. What is the normalised P/E?

- A 25.0x
- B 22.5x
- C **27.8x**
- D 20.8x

Ans: C**WORKING**

Normalised PAT = $120 - 30 = ₹90 \text{ cr}$ EPS = $90 \div 10 = ₹9.00$

Normalised P/E = $250 \div 9 = 27.8\times$ (reported P/E = $250 \div 12 = 20.8\times$)

△ A gain on sale of assets is not recurring — by his 2x2 it is ignored entirely.
The stock is a third more expensive than the reported number suggests.

Q8

DCF

EBIT ₹300m in perpetuity, tax 30%, cost of equity 18%, debt ₹200m at 10%, zero growth. What is the equity value by the FCFE route?

- A ₹980.0m
- B **₹1,088.9m**
- C ₹1,166.7m
- D ₹1,288.9m

Ans: B**WORKING**

Interest = $200 \times 10\% = 20$

PAT = $(300 - 20) \times 0.7 = 196$ ($g = 0$, so FCFE = PAT)

Equity = $196 \div 0.18 = ₹1,088.9\text{m}$

EV = $1,088.9 + 200 = ₹1,288.9\text{m}$

Note the order: FCFE needs no weights, so equity comes first — and only then can you build market-value weights for the WACC. That is the escape from circularity.

Q9

DCF

For the same firm, what is the capital cash flow and the rate it must be discounted at?

- A CCF = 216 at a WACC of 16.29%
- B CCF = 216 at a pre-tax WACC of 16.76%**
- C CCF = 210 at a WACC of 16.29%
- D CCF = 210 at a pre-tax WACC of 16.76%

Ans: B

WORKING

$$FCF = 300 \times 0.7 = 210$$

$$CCF = FCF + t \cdot K_d \cdot D = 210 + (0.30 \times 20) = \mathbf{216}$$

$$\text{pre-tax WACC} = (1088.9/1288.9)(18\%) + (200/1288.9)(10\%) = \mathbf{16.759\%}$$

$$\text{check: } 216 \div 0.16759 = \mathbf{1,288.9} \checkmark \text{ same EV}$$

⚠ CCF already contains the tax shield, so the discount rate must NOT carry the $(1-t)$. Using the after-tax WACC of 16.29% counts the shield twice and overstates value.

Q10

DCF

A levered firm has $K_e = 20\%$, $K_d = 10\%$, $D/E = 0.5$ and a tax rate of 30%. What is p , the unlevered cost of equity?

- A 15.00%
- B 16.67%
- C 18.18%
- D 17.41%**

Ans: D

WORKING

$$K_e = p + (p - K_d)(1 - t)(D/E)$$

$$20 = p + (p - 10)(0.7)(0.5) = p + 0.35p - 3.5$$

$$23.5 = 1.35p \Rightarrow p = \mathbf{17.407\%}$$

p is pure business risk — what shareholders would demand with no debt at all. It is the starting point for every APV valuation.

Q11

DCF

Year-5 free cash flow is ₹240m, terminal growth 5%, WACC 12%. What is the present value of the terminal value today?

- A ₹3,600.0m
- B ₹1,823.9m
- C ₹2,042.7m
- D ₹2,144.9m

Ans: C

WORKING

TV at end of yr 5 = $240 \times 1.05 \div (0.12 - 0.05) = 252 \div 0.07 = ₹3,600\text{m}$

PV = $3,600 \div 1.12^5 = 3,600 \div 1.76234 = ₹2,042.7\text{m}$

△ Two slips live here. ₹1,823.9m drops the $(1+g)$; discounting six periods instead of five gives ₹1,823.8m as well. The TV sits at the end of year 5, so it comes back exactly five periods.

Q12

DCF

Your FCF model gives an equity value of ₹500 cr and your FCFE model ₹460 cr. The model uses a constant WACC, growth of 6%, and holds debt fixed at ₹200 cr. What is the most likely cause?

- A The tax rate has been applied to PBT instead of EBIT
- B The terminal value has been discounted one period too many
- C The beta was estimated over the wrong window
- D Net new borrowing of $g \times D = ₹12$ cr a year is missing from FCFE

Ans: D

WORKING

constant WACC $\Rightarrow$ constant D/E $\Rightarrow$ debt must grow at g

net new borrowing = $g \times D = 0.06 \times 200 = ₹12$ cr each year

FCFE = FCF - Interest(1-t) + net new borrowing

Nine times out of ten this is the cause of an FCF/FCFE mismatch. Either let the debt grow at g and keep the rates constant, or hold debt fixed and switch to APV.

Q13

DCF

Enterprise value is ₹1,000 cr. The firm has ₹200 cr of debt, ₹100 cr of preference capital, ₹50 cr of surplus cash excluded from the cash flows, and a contingent liability with an expected value of ₹20 cr. What is the equity value?

- A ₹730 cr
- B ₹750 cr
- C ₹700 cr
- D ₹630 cr

Ans: A

WORKING

$$1,000 + 50 \text{ surplus cash} - 200 \text{ debt} - 100 \text{ preference} - 20 \text{ contingent} \\ = ₹730 \text{ cr}$$

⚠ Preference capital is a claim ahead of equity — it goes into the WACC at its own cost with no $(1-t)$, and it must come out at the bridge. Miss it and you hand the preference holders' money to the equity.

Q14

Terminal value

A terminal year shows EBITDA of \$250 million and free cash flow of \$100 million. An exit multiple of 8x EBITDA is applied and terminal growth is 4%. What cost of capital is implied by that exit multiple?

- A 9.0%
- B 9.2%**
- C 5.2%
- D 13.2%

Ans: B

WORKING

$$TV = 8 \times 250 = \$2,000m$$

$$TV = FCF(1+g) \div (WACC - g) \implies 2,000 = 104 \div (WACC - 0.04)$$

$$WACC - 0.04 = 104 \div 2,000 = 0.052 \implies WACC = 9.2\%$$

The mirror of the PYQ: there you were given the WACC and asked for the multiple; here you are given the multiple and asked for the rate. Same identity, rearranged. If the WACC your model actually uses is 13%, then an 8x exit multiple is too generous.

Q15

Cost of capital

A company has migrated to section 115BAA. Its pre-tax cost of debt is 9.5%. What is the after-tax cost of debt?

- A 6.18%
- B 9.50%
- C **7.11%**
- D 7.13%

Ans: C

WORKING

§115BAA effective rate = 22% + 10% surcharge + 4% cess = **25.168%**

$$K_d(1 - t) = 9.5\% \times (1 - 0.25168) = \mathbf{7.109\%}$$

7.13% is the answer from a rounded 25%; 6.18% uses the old regime's 34.94%. Check the regime in the tax note of the annual report — never assume it.

Q16

Beta

The covariance between a stock's returns and the index is 0.0288 and the variance of the index is 0.0225. What is the stock's beta?

- A 0.64
- B **1.28**
- C 0.78
- D 1.20

Ans: B

WORKING

$$\beta = \text{Cov}(r_i, r_m) \div \text{Var}(r_m) = 0.0288 \div 0.0225 = \mathbf{1.28}$$

Equivalently $\beta = \rho \times (\sigma_i \div \sigma_m)$. Here $\sigma_m = \sqrt{0.0225} = 15\%$. If the stock's σ were 24%, then $\rho = 1.28 \times 15/24 = 0.80$ and $R^2 = 0.64$.

Q17

Beta

A comparable company has a levered beta of 1.30, a debt-to-equity ratio of 0.80 and a tax rate of 25%. What is its unlevered beta?

- A 1.04
- B 0.81**
- C 0.72
- D 0.87

Ans: B

WORKING

$$\begin{aligned}\beta_U &= \beta_L \div [1 + (1 - t)(D/E)] \\ &= 1.30 \div [1 + 0.75 \times 0.80] = 1.30 \div 1.60 = \mathbf{0.8125}\end{aligned}$$

Unlever each peer with its own D/E and tax rate, take the median (robust to outliers in a small sample), then relever at your company's target D/E.

Q18

Beta

A company makes a 1-for-4 rights issue at ₹200 when the cum-rights price is ₹300. What price should the return series use on the ex-rights day?

- A ₹250
- B ₹260
- C ₹275
- D ₹280**

Ans: D

WORKING

$$\text{TERP} = (4 \times 300 + 1 \times 200) \div 5 = (1,200 + 200) \div 5 = \mathbf{₹280}$$

△ The price 'falling' from ₹300 to ₹280 is not a -6.7% return — it is the arithmetic of issuing cheap shares. Unadjusted, that day enters the beta regression as a spurious negative return.

Q19

Cost of capital

The spot rate is ₹85/\$, the dollar interest rate is 2% and the rupee rate is 6.5%. What is the one-year forward rate under interest rate parity?

- A ₹81.41/\$
- B ₹89.25/\$
- C **₹88.75/\$**
- D ₹85.00/\$

Ans: C

WORKING

$$\begin{aligned}\text{Forward} &= \text{Spot} \times (1 + r_{\text{₹}}) \div (1 + r_{\$}) \\ &= 85 \times 1.065 \div 1.02 = 85 \times 1.04412 = \mathbf{₹88.75}\end{aligned}$$

This is why a dollar loan at 6% is not 6% debt for an Indian borrower: convert the whole schedule at expected forwards and take the rupee IRR. In his worked case a 6% dollar loan came out at a rupee cost of **12.30%**.

Q20

Cash flow

NOPAT ₹400m, depreciation ₹120m, capex ₹260m, increase in operating working capital ₹60m, increase in other non-interest-bearing liabilities ₹15m. What is free cash flow?

- A ₹200m
- B ₹185m
- C ₹245m
- D **₹215m**

Ans: D

WORKING

$$\begin{aligned}\text{Gross cash flow} &= 400 + 120 &&= \mathbf{520} \\ \text{Gross investment} &= 260 + 60 - 15 &&= \mathbf{305} \\ \text{Free cash flow} &= 520 - 305 &&= \mathbf{₹215m}\end{aligned}$$

An increase in non-interest-bearing liabilities is a source of funds, so it reduces the investment the business needs.

Q21

ROIC

NOPAT for FY25 is ₹375 cr. Operating invested capital was ₹2,500 cr at the end of FY24 and ₹2,800 cr at the end of FY25. What is ROIC for FY25?

- A 15.0%
- B 13.4%
- C 14.2%
- D 11.9%

Ans: A

WORKING

$$\text{ROIC} = \text{NOPAT}_t \div \text{OIC}_{(t-1)} = 375 \div 2,500 = 15.0\%$$

△ Mind the lag. This year's profit was earned on last year's capital. Using the closing figure gives 13.4%, and using the average gives 14.2% — both understate the return the business actually earned on the capital it had.

Q22

Terminal value

An analyst's terminal year shows NOPAT ₹600m with net investment of zero, ROIC 14%, WACC 11%, and a terminal value of ₹7,500m from an exit multiple. What is the corrected terminal value?

- A ₹7,500.0m
- B ₹6,011.9m
- C ₹5,833.3m
- D ₹6,500.0m

Ans: B

WORKING

mistake: NI = 0 forces $g = 0$, but the exit multiple implies positive g

$$g = (\text{TV} \times \text{WACC} - \text{FCF}) \div (\text{TV} + \text{FCF}) = (825 - 600) \div 8,100 = 2.778\%$$

$$\text{IR} = g \div \text{ROIC} = 2.778 \div 14 = 19.84\%$$

$$\text{NI} = 600 \times 19.84\% = 119.05 \quad \text{FCF} = 600 - 119.05 = 480.95$$

$$\text{TV} = 480.95 \times 1.02778 \div (0.11 - 0.02778) = ₹6,011.9\text{m}$$

△ The naive figure overstates terminal value by ₹1,488m — about 25% — because it credits growth the company never paid for.

Q23

VC method

A fund invests \$10m and requires 40% a year for five years. Exit-year earnings are projected at \$12m and comparables trade at 15x. What share must the fund own at exit?

- A 35.9%
- B 22.4%
- C **29.9%**
- D 25.0%

Ans: C

WORKING

Terminal value = $15 \times 12 = \$180\text{m}$

FV of investment = $10 \times 1.40^5 = 10 \times 5.3782 = \53.78m

Required share = $53.78 \div 180 = 29.88\%$

Q24

VC method

The fund now insists management must own 12% of the company by exit. What share must it require today?

- A 29.88%
- B **33.95%**
- C 41.88%
- D 26.29%

Ans: B

WORKING

required today = required at exit $\div$ (1 - dilution)

= $29.88\% \div (1 - 0.12) = 33.95\%$

⚠ Dilution is multiplicative, not additive. $29.88 + 12 = 41.88\%$ is the trap. The option pool is funded entirely out of the founders' stake.

Q25

VC method

The company has 4m shares before the round. The fund takes 33.95% for \$10m. What is the price per share?

A \$7.36

B **\$4.86**

C \$5.87

D \$2.94

Ans: B

WORKING

$$\text{New shares} = 4\text{m} \times [0.3395 \div (1 - 0.3395)] = 4\text{m} \times 0.5140 = \mathbf{2.056\text{m}}$$

$$\text{Price} = \$10\text{m} \div 2.056\text{m} = \mathbf{\$4.864}$$

$$\text{check: post} = 10 \div 0.3395 = 29.46 \implies \text{pre} = 19.46 \implies 19.46 \div 4\text{m} = \mathbf{\$4.86} \checkmark$$

Without the option pool the price would be \$5.87 — the pool costs the founders about a rupee in every six of share price.

Q26

Anti-dilution

Series A holds 8m shares with a conversion price of \$2.00. Total shares outstanding before the new round are 20m. The company raises \$6m by issuing 5m new shares. Under weighted-average anti-dilution, what is Series A's new conversion price?

- A \$1.53
- B \$2.00
- C \$1.20
- D **\$1.84**

Ans: D

WORKING

$$\begin{aligned}\text{NCP} &= [(OB \times OCP) + \text{New\$}] \div OA \\ &= [(20m \times \$2.00) + \$6m] \div (20m + 5m) \\ &= (\$40m + \$6m) \div 25m = \mathbf{\$1.84}\end{aligned}$$

Series A now converts into $8m \times (2.00 \div 1.84) = 8.70m$ shares. The new investor's percentage is unchanged — the protection transfers value from the founders to the earlier investor. A full ratchet would reset the price all the way to the new round's price.

Q27

VC method

Founders hold 2,000,000 shares and Series A holds 500,000 (20% of 2.5m). At Series B the pre-money is \$60 million and a new investor puts in \$15 million. How much must Series A invest to stay at 20%?

- A \$3.00 million
- B \$4.50 million
- C **\$3.75 million**
- D \$15.00 million

Ans: C

WORKING

Price = $\$60\text{m} \div 2.5\text{m} = \24.00 A's stake = $500,000 \times 24 = \$12\text{m}$

$$(12 + x) \div (60 + 15 + x) = 0.20$$

$$12 + x = 15 + 0.2x \implies 0.8x = 3 \implies x = \$3.75\text{m}$$

Post-money = \$78.75m; A holds \$15.75m = 20.0% ✓. \$3.00m is the trap of taking 20% of the \$75m post-money and subtracting the \$12m already held — it forgets that your own cheque raises the post-money.

Q28

Private valuation

An owner-manager draws ₹6 lakh a year; the market salary for the role is ₹30 lakh. Reported PAT is ₹2.40 cr and the tax rate is 25%. What is normalised PAT?

- A ₹2.16 cr
- B ₹2.40 cr
- C ₹1.96 cr
- D ₹2.22 cr

Ans: D

WORKING

Salary shortfall = $30 - 6 = ₹24 \text{ lakh} = ₹0.24 \text{ cr}$ (pre-tax)

After-tax effect = $0.24 \times (1 - 0.25) = ₹0.18 \text{ cr}$

Normalised PAT = $2.40 - 0.18 = ₹2.22 \text{ cr}$

Normalisation runs both ways: owners often underpay themselves to flatter profit before a sale, just as often as they overpay to reduce tax. Also strip personal expenses, ghost-employee salaries and non-recurring items.

Q29

Private valuation

A comparable has a beta of 0.90 and its regression against the index has an R^2 of 0.36. With a risk-free rate of 7% and an equity risk premium of 7%, what cost of equity applies to a completely undiversified owner?

- A 17.5%
- B 13.3%
- C 24.5%
- D 10.5%

Ans: A

WORKING

$$\rho = \sqrt{R^2} = \sqrt{0.36} = 0.60$$

$$\text{Total beta} = \beta \div \rho = 0.90 \div 0.60 = 1.50$$

$$K_e = 7\% + 1.50 \times 7\% = 17.5\% \quad (\text{diversified: } 7 + 0.9 \times 7 = 13.3\%)$$

Total beta is the rigorous version of the 'double the beta' shortcut. An owner holding only this asset bears the whole σ , not just the systematic part.

Q30

Private valuation

A 100% control stake in a private company is worth ₹300 cr. You are valuing a **20% minority** stake. Applying a 20% minority discount and a 25% DLOM, what is that stake worth?

A ₹48 cr

B ₹36 cr

C ₹45 cr

D ₹60 cr

Ans: B

WORKING

Pro-rata share = $300 \times 20\%$ = ₹60 cr

Less minority disc. = $60 \times (1 - 0.20)$ = ₹48 cr

Less DLOM = $48 \times (1 - 0.25)$ = ₹36 cr

△ The discounts are applied in sequence, not added (a combined 45% haircut would give ₹33 cr). And a DCF built on management's forecasts is already a control value — that is why the minority discount is needed at all.

Q31

A private company is about to list. Which two things should change in the valuation?

- A Only the capital structure changes
- B Nothing changes — the cash flows are the same
- C **The DLOM disappears and K_e moves toward the plain CAPM figure**
- D The DLOM rises and K_e falls

Ans: C

WORKING

shares become marketable $\Rightarrow$ illiquidity discount goes

IPO investors are diversified $\Rightarrow K_e \rightarrow$ plain CAPM

both effects push value **up**

In the Kavita case that is the move from ₹259 cr to something approaching the undiscounted ₹346 cr, and then higher again as K_e falls from 20% toward 11.84%.

Watch the use of proceeds: taken out by the owners $\rightarrow$ ignore; used to repay debt $\rightarrow$ the debt ratio and cost of capital change; held for planned reinvestment $\rightarrow$ add to the DCF.

Q32

Real options

A patent gives exclusive rights for 8 years. Developing the drug costs ₹800 cr today and the PV of the cash flows if developed is ₹700 cr. Outcomes are highly uncertain. Is the patent worth nothing?

- A It has value only once the NPV turns positive
- B No — the right to develop later is a call option with positive value despite the negative NPV**
- C Yes — the NPV is -₹100 cr so the project should be abandoned
- D It is worth exactly -₹100 cr

Ans: B

WORKING

Map onto a call: $S = 700$ $K = 800$ $T = 8 \text{ yrs}$ $\sigma = \text{high}$

Out of the money today, but **time value** > 0 whenever $\sigma > 0$

All three conditions hold: flexibility (a right, not an obligation), uncertainty (with no uncertainty an option is worth only its intrinsic value), and exclusivity (the patent). A generic 'option to expand the factory' usually fails the third.

Q33

Real options

In Laura Martin's stealth-tier valuation, the market value per home passed is \$3,000 and 120 channels are currently lit. What value does she assign to the underlying, S ?

- A \$17.65
- B \$25.00**
- C \$23.15
- D \$250.00

Ans: B**WORKING**

$S = \text{value per home passed} \div \text{lit channels} = 3,000 \div 120 = \25.00

(in the case itself: $2,500 \div 108 = \$23.15$)

△ Know what to criticise: she takes $\sigma = 50\%$ from a one-month option on Cox equity to price a ten-year option on a channel's cash flows — wrong horizon and levered rather than asset volatility. And if the DCF's terminal value already grows on digital services, adding the option double-counts.

Q34

AI valuation

A private company is valued at \$120bn. Its most recent month's revenue was \$500m. What is its run-rate revenue multiple, and what is the caution?

- A 24x — using the trailing twelve months
- B 240x — the multiple on monthly revenue
- C 10x — half the annualised figure to be conservative
- D 20x — run-rate annualises the latest month and is not trailing revenue**

Ans: D**WORKING**

Run-rate revenue = $500\text{m} \times 12 = \6bn

Multiple = $120 \div 6 = 20\text{x}$

△ In his Anthropic example the same \$965bn valuation gave 50x on Q1 annualised, 31x on H1 annualised and 20x on run-rate. In a fast-growing company the choice of denominator matters more than any argument about the numerator — quoting a multiple without stating the basis is meaningless.

EV/EBITDA is regressed on ROIC for 12 companies: $EV/EBITDA = 4.2 + 0.35 \times ROIC(\%)$, with t-statistics of 2.9 and 2.6 and an R^2 of 41%. What is the best interpretation?

- A ROIC fully determines EV/EBITDA in this sector
- B The slope is insignificant, so ROIC does not affect the multiple
- C Both coefficients are significant, and ROIC explains 41% of the cross-sectional variation — the other 59% is everything else**
- D 41% of the companies in the sample are correctly valued

Ans: C

WORKING

$n = 12, k = 2 \Rightarrow df = 10$, 5% critical $t \approx 2.23$

2.9 and 2.6 both exceed it $\Rightarrow$ both coefficients significant

$R^2 = 41\% \Rightarrow$ 41% of the VARIATION explained, not 41% of the companies

Contrast with the P/E PYQ, where the slope's t of 0.8 killed it. Here the relationship is real — which is exactly the logic behind Laura Martin's ROIC target price, where the same regression on cable companies had $R^2 = 70\%$.

⚠ R^2 is about variance explained. It never says anything about how many companies are "correctly valued", and a significant slope never means a variable is the only thing that matters.

An Indian private company issues shares to a resident investor at a premium. Under which statute does the valuation requirement arise, and by what name is it known?

- A The Companies Act — preferential allotment
- B SEBI regulations — open offer
- C FEMA — transfer to a non-resident
- D **The Income-tax Act — angel tax**

Ans: D

WORKING

Income-tax Act: an amount above ₹50,000 received without consideration is taxable.

Applied to a share premium, it taxes the **excess of issue price over fair value**.

The rest of the map: Companies Act — preferential allotment, buyback, merger/demerger. SEBI — open offer, preferential allotment. FEMA — overseas direct investment and resident ↔ non-resident transfers. IBC — liquidation. Ind AS — purchase price allocation, options, financial instruments. Also worth knowing: §9(1)(i), the retrospective provision behind the Vodafone dispute — 'tax anything, anytime'.

Q37

Guest lecture

A registered valuer uses a risk-free rate of 7%, a beta of 0.93, a market risk premium of 7%, and adds a company-specific risk premium of 3.5%. What cost of equity does she report?

- A 17.01%
- B 13.51%
- C 20.51%
- D 16.51%

Ans: A

WORKING

$$\text{CAPM part} = 7\% + 0.93 \times 7\% = 7\% + 6.51\% = 13.51\%$$

$$K_e = 13.51\% + 3.5\% \text{ CSRP} = 17.01\%$$

CSRP compensates for risks beta cannot see: size, client concentration, promoter dependency, single-location risk, key-person exposure. Indian practice caps it near 4% and runs the market risk premium at 6–7%. It is the practitioner's route to the same problem total beta solves academically.

Q38

Guest lecture

A target's trade creditors are on 60-day terms but it is actually paying at 165 days. How should this be handled in the bridge from enterprise value to equity value?

- A Leave it entirely inside working capital
- B The excess is a stretch creditor — a debt-like item to be subtracted**
- C Add it as a surplus asset
- D Ignore it, since trade credit carries no interest

Ans: B

WORKING

normal terms = 60 days → stays in operating working capital

actual = 165 days

excess 105 days → disguised borrowing → debt-like item, SUBTRACT

The full debt-like list: borrowings · unfunded gratuity and leave encashment · stretch creditors · non-convertible preference shares · lease liabilities · contingent liabilities at estimated impact × probability · minority interest.

A database records a transaction at $6\times$ EV/EBITDA. The advisor knows the deal was actually negotiated on net asset value plus two years' PAT as goodwill. What is the lesson for comparable transaction multiples?

- A NAV-based deals cannot be compared at all
- B The multiple is fine because the price paid is the price paid
- C The multiple was back-solved from a price set on a different basis, so CTM data can be spurious**
- D The database is simply wrong and should be corrected

Ans: C

WORKING

deal negotiated on NAV + goodwill → database stores a **multiple**
⇒ the $6\times$ describes nothing about how the price was actually reached

The other database traps: a deal where 100% of equity was bought but only 60% of the consideration was disclosed; period adjustments to revenue and margins; management replacement cost; actuarial gratuity provisioning — all inside a real 'adjusted EBITDA' and none of them visible in the database. Different databases also compute the same company's multiple differently.

Q40

Guest lecture

A business is midway through a heavy capital-expenditure programme and is growing at 22%, well above its long-run rate. Which terminal-value method is most appropriate?

- A A zero-growth perpetuity
- B The H-model, in which growth fades from the horizon rate to the perpetual rate**
- C Gordon growth with a higher terminal g to reflect the capex
- D An exit multiple applied to current-year EBITDA

Ans: B

WORKING

Gordon assumes growth **snaps** from 22% to ~4% in a single year – not credible
H-model: growth **fades linearly** to the perpetual rate, then continues forever
Three terminal-value methods are on the syllabus: Gordon growth, the H-model, and exit multiples (EV/EBITDA or EV/EBIT on terminal-year earnings). The alternative to the H-model is simply to lengthen the explicit forecast period until steady state is credible — which is the same idea by another route.